Marked groups, kernels and intersections

Abstract

Many constructions of incidence geometries require the same two tasks: control the intersections of subgroups generated by selected markers, and identify the permutation group acting on the resulting objects. We give a common treatment in which a quotient supplies the visible geometry and its kernel records the remaining intersection obstructions. The main construction assembles local extensions over specified sets of markers. Injectivity on every proper local parabolic is enough to isolate the entire local kernels, even when their supports are nested. It follows that the full fiber product is generated by the prescribed markers and has an exact kernel profile on every parabolic. We derive local recognition, deletion maps, lower-central identities and separator factorizations from the same proof. Exact coordinate profiles also transfer typed coset Helly; with the necessary involutory local markings this gives regular geometries. A finite nilpotent quotient and an invariant subgroup ladder then give a complete parameter space for marked finite two-group quotients with a prescribed class-two quotient. Further results treat exact quotient intersections, comparison through one exceptional parabolic, faithful actions on retained objects, relative generation and Frattini coordinates. These results are intended to be cited with explicit hypothesis bindings; local presentations, geometric identifications and finite classifications remain the responsibility of their applications.

1. Markings, images and kernels

A marked group is a group G together with a finite indexed generating family $(r_i)_{i\in I}$. Except where stated, the markers need not be involutions or distinct. For $S \subseteq I$, put

$$G_S = \langle r_i : i \in S\rangle, \qquad G_\emptyset = 1.$$

We call these subgroups parabolics, retaining their index sets even when two of them happen to coincide. The marking has the intersection property, or IP, if

$$G_S \cap G_T = G_{S\cap T} \quad (S, T \subseteq I). \tag{1.1}$$

A marked homomorphism sends each source marker to the corresponding target marker. Its image on G_S is therefore the target parabolic with index set S.

All subgroup products below are products of sets unless a subgroup assertion is made. We write $\gamma_1(G) = G$ and $\gamma_{j+1}(G) = [\gamma_j(G), G]$. We use the elementary correspondence and isomorphism theorems for groups. In the finite p-group arguments we retain Burnside's basis theorem, the nilpotence of finite p-groups, and the usual multilinear commutator description of the lower-central graded Lie ring. Precisely, its degree-one part generates the graded Lie ring, and degree j is generated by left-normed commutators of weight j in any generating set. In Section 8 we explicitly retain the normal-form theorem for an amalgamated free product. None of these classical inputs is presented as a result originating in this manuscript.

Lemma U.K0 (lifting one intersection). Let $q : H \to Q$ be a homomorphism and let $C \le A \cap B$, where $A, B, C \le H$. If

$$q(A) \cap q(B) = q(C),$$

then

$$A \cap B = C \quad \Longleftrightarrow \quad A \cap B \cap \ker q \le C. \tag{1.2}$$

Proof. The forward implication is immediate. For the converse, let $x \in A \cap B$. Its image belongs to $q(C)$, so choose $c \in C$ with $q(c) = q(x)$. Then $xc^{-1} \in A \cap B \cap \ker q$, which is contained in C by hypothesis. Multiplying by c gives $x \in C$. The inclusion $C \le A \cap B$ gives equality. □

Theorem U.K (kernel profiles). Let $q : H \twoheadrightarrow Q$ be a marked epimorphism and suppose that the marking of Q has IP. Write $K = \ker q$ and $K_S = K \cap H_S$. Then H has IP if and only if

$$K_S \cap K_T = K_{S\cap T} \quad (S, T \subseteq I). \tag{1.3}$$

Proof. Apply U.K0 to $A = H_S$, $B = H_T$, and $C = H_{S\cap T}$. The image hypothesis is exactly IP in Q. The subgroup on the left of the containment in (1.2) is $K_S \cap K_T$. Its containment in $H_{S\cap T}$ is equivalent to its containment in $K_{S\cap T}$; the reverse inclusion always holds. This proves each required intersection and its converse. □

The next result concerns the opposite direction. Passing to a quotient can create new intersections, and this defect is a product condition in its kernel.

Theorem U.Q (exact intersections of images). Let $q : H \to Q$ be a homomorphism with kernel N, and let $A, B \leq H$. Then

$$q(A) \cap q(B) = q(A \cap B) \quad \iff \quad N \cap AB = (N \cap A)(N \cap B). \tag{1.4}$$

If q is injective on A and B, the right side reduces to $N \cap AB = \{1\}$. If $N \leq A$, the equality always holds.

Proof. Suppose the image intersection is exact, and take $ab \in N$, where $a \in A$ and $b \in B$. Since $q(a) = q(b)^{-1}$, choose $c \in A \cap B$ with this image. Then

$$ab = (ac^{-1})(cb) \in (N \cap A)(N \cap B).$$

The reverse inclusion requires no hypothesis. Conversely, if $q(a) = q(b)$, then $ab^{-1} \in N \cap AB$. Write $ab^{-1} = uv$, with $u \in N \cap A$ and $v \in N \cap B$. The element $u^{-1}a = vb$ lies in $A \cap B$ and has image $q(a)$, proving the required image equality. Under injectivity both factors on the right of (1.4) are trivial. If $N \leq A$, both sides of that equality are N. □

Proposition U.Coset (exact coset transport). Let $(H_i)_{i\in I}$ be subgroups of G, let $B = \bigcap_i H_i$, and let $N \trianglelefteq G$ satisfy $N \leq B$. Quotienting by N gives a bijection

$$H_i g \longmapsto (H_i/N)(gN) \tag{1.5}$$

on the typed right cosets. It preserves the intersections of arbitrary families of those cosets, including the existence of a common element. It consequently preserves incidence, flags, residues and all properties determined by the resulting incidence system. The actions of G and G/N have the same permutation image. Maximality of any inclusion between intersections of the H_i is preserved.

Proof. The inverse image of the coset on the right of (1.5) is exactly $H_i g$. Thus two source cosets give the same target coset precisely when they were equal, and taking inverse images commutes with all intersections. Surjectivity of the quotient map supplies a source representative of each target coset and a common lift whenever a target intersection is nonempty. This proves the assertions about incidence and common elements. The quotient action is visibly the same action on the identified coset set. Every subgroup in any interval under consideration contains N; the subgroup correspondence therefore preserves that interval and its maximal inclusions. □

2. Deletion maps and supported extensions

A deletion for $S \subseteq I$ is an endomorphism $e_S : G \to G$ satisfying $e_S(r_i) = r_i$ for $i \in S$ and $e_S(r_i) = 1$ otherwise. Requiring these values matters: a general retraction onto G_S need not be a deletion.

Proposition U.D (Boolean deletions). If a marked group has a deletion e_S for every $S \subseteq I$, then

$$e_S e_T = e_{S\cap T}, \qquad \operatorname{im} e_S = \operatorname{Fix}(e_S) = G_S. \tag{2.1}$$

The marking has IP, and

$$\ker e_S = \langle\langle r_i : i \notin S \rangle\rangle_G, \qquad G = (\ker e_S) \rtimes G_S. \tag{2.2}$$

Proof. The two maps in the first equality agree on every generator. The image is generated by precisely the retained markers, and each of those markers is fixed; this proves the image and fixed-point statements. An element of $G_S \cap G_T$ is fixed by $e_S e_T = e_{S\cap T}$, so belongs to $G_{S\cap T}$. The reverse inclusion is immediate.

Let M be the normal closure in (2.2). Clearly $M \le \ker e_S$. In G/M, every omitted marker is trivial, so the image of any word equals the image of its deletion. A word in $\ker e_S$ is therefore trivial in G/M, proving equality. Finally, for $g \in G$, the factorization $g = (g e_S(g)^{-1}) e_S(g)$ has factors in $\ker e_S$ and G_S. Their intersection is trivial because e_S fixes G_S. This is the asserted semidirect product. □

We now assemble extensions whose extra kernel is invisible on each proper part of its support. The groups and kernels may be infinite and nonabelian.

Theorem U.SK (supported-kernel realization). Let I be finite and let $\mathcal{F}$ be a finite family of distinct subsets of I. Let $B = \langle b_i : i \in I \rangle$ have IP. For every $S \in \mathcal{F}$, suppose the following data are given:

1. A deletion homomorphism $p_S : B \to B_S$, fixing b_i for $i \in S$ and killing b_i otherwise.
2. A group $E_S = \langle e_{S,i} : i \in S \rangle$ and a marked epimorphism $q_S : E_S \to B_S$.
3. Injectivity of q_S on $(E_S)_T = \langle e_{S,i} : i \in T \rangle$ for every $T \subsetneq S$.

Put $N_S = \ker q_S$, and form the group

$$H = \left\{ (b, (x_S)_{S\in\mathcal{F}}) \in B \times \prod_{S\in\mathcal{F}} E_S : q_S(x_S) = p_S(b) \text{ for every } S \right\}. \tag{2.3}$$

For $i \in I$, define h_i to have base coordinate b_i and S-coordinate $e_{S,i}$ when $i \in S$, and 1 otherwise. These elements generate all of H. If $\pi : H \to B$ is the base projection and $H_L = \langle h_i : i \in L \rangle$, then, for every $L \subseteq I$,

$$\pi(H_L) = B_L, \qquad \ker\pi = \prod_{S\in\mathcal{F}} N_S, \qquad \ker\pi \cap H_L = \prod_{\substack{S\in\mathcal{F}\\ S\subseteq L}} N_S. \tag{2.4}$$

The factors on the right are embedded as pure coordinate subgroups. The marking of H has IP. If B and all E_S are finite, then

$$|H_L| = |B_L| \prod_{\substack{S\in\mathcal{F}\\ S\subseteq L}} |N_S|. \tag{2.5}$$

Proof. The compatibility equations in (2.3) are preserved by products and inverses, so H is a subgroup of the displayed direct product. The prescribed tuple h_i satisfies those equations by the literal deletion rule. Let H' be the subgroup generated by these tuples. Its base image is B, and the projection of $H'_S = \langle h_i : i \in S \rangle$ onto E_S is onto.

We first prove, by induction on inclusion among the finitely many supports, that every pure coordinate element of N_S belongs to H'_S. Fix $n \in N_S$. Choose a word on the markers indexed by S which evaluates to n in E_S, and evaluate the same word in H', obtaining $w \in H'_S$. Its base coordinate lies in B_S and is $q_S(n) = 1$.

Consider its coordinate at another support T. Because the word uses only markers from S, that coordinate lies in $(E_T)_{S\cap T}$. Its image under q_T is $p_T(1) = 1$. If $T \not\subseteq S$, then $S \cap T \subsetneq T$, and the proper-parabolic injectivity hypothesis forces this coordinate to be 1. The only possible nonidentity coordinates of w are consequently at supports $T \subseteq S$. Its S-coordinate is the prescribed n, while every coordinate at a proper support $T \subsetneq S$ belongs to N_T.

For each such proper support, the induction hypothesis supplies the pure coordinate element having exactly that value in N_T, as an element of $H'_T \leq H'_S$. Multiply w by the inverses of those elements. Distinct pure coordinate subgroups commute in the ambient direct product, so these operations remove the unwanted coordinates without changing the S-coordinate. The result is the pure element n in coordinate S, and remains in H'_S. A minimal support has no smaller coordinates to remove, so this argument includes the initial step. No commutativity within an individual kernel has been used.

The kernel of π in the full group H is exactly the direct product of the N_S, by (2.3). The induction places that entire kernel in H'. Since H' already maps onto B, every $h \in H$ has the same base coordinate as some $h' \in H'$; then $h(h')^{-1} \in \ker \pi \leq H'$. Thus $H' = H$.

The image equality $\pi(H_L) = B_L$ follows on generators. If $x \in H_L \cap \ker \pi$, its S-coordinate lies in $(E_S)_{L \cap S}$ and in N_S. This coordinate is trivial whenever $S \not\subseteq L$, again by proper-parabolic injectivity. Conversely, when $S \subseteq L$, the isolation argument placed the entire pure N_S in $H_S \leq H_L$. This proves all three assertions in (2.4).

The coordinate products in (2.4) satisfy

$$\left(\prod_{S \subseteq L} N_S\right) \cap \left(\prod_{S \subseteq M} N_S\right) = \prod_{S \subseteq L \cap M} N_S.$$

Theorem U.K now proves IP in H. More explicitly, an element of $H_L \cap H_M$ has base in $B_{L \cap M}$; subtracting a lift from $H_{L \cap M}$ leaves an element in both displayed kernel products, hence in the kernel product for $L \cap M$. This also shows exactly where IP of the base is used. Finally (2.5) is the order formula for the surjection $H_L \to B_L$ with the kernel just computed. □

Corollary U.SK.Local (recognition on an antichain). Under U.SK, if $\mathcal{F}$ is an antichain under inclusion, then the coordinate projection $H_S \to E_S$ is a marked isomorphism for every $S \in \mathcal{F}$.

Proof. It is onto on generators. If an element maps to 1, its base coordinate also vanishes, since the base value of a word on S is the q_S-image of its S-coordinate. Formula (2.4) says that the remaining kernel in H_S has only the factor N_S: the antichain excludes every other support contained in S. The S-coordinate detects that factor, so the element is 1. □

Corollary U.SK.Delete (derived deletions). If, in addition, B has deletions for every subset of I, then every E_S and H has all its Boolean deletions.

Proof. For $T \subsetneq S$, the restriction $q_S : (E_S)_T \to B_T$ is a marked isomorphism: it is onto on generators and injective by hypothesis. Compose q_S, the deletion $B_S \to B_T$, and the inverse of this isomorphism. This gives a homomorphism $E_S \to (E_S)_T$ that fixes exactly the T-markers and kills the others. For $T = S$ use the identity. Proposition U.D supplies the Boolean identities.

For $L \subseteq I$, apply the base deletion e_L and, in coordinate S, the local deletion onto $S \cap L$. The two sides of each compatibility equation agree on the local markers; the base deletions commute by U.D. Thus the coordinatewise operation preserves (2.3). On h_i it fixes h_i if $i \in L$, and kills it otherwise. It is the required deletion of H. □

Corollary U.SK.Central (lower-central kernel). Under U.SK, suppose $c \geq 1$, $\gamma_c(B) = 1$, and $N_S = \gamma_c(E_S)$ for every support. Then

$$\gamma_c(H) = \ker \pi. \tag{2.6}$$

Proof. Applying π to a commutator of weight c gives 1, so $\gamma_c(H) \leq \ker \pi$. For the converse, repeat the inclusion induction in the proof of U.SK. The projection $H_S \to E_S$ is onto; therefore its image on $\gamma_c(H_S)$ is $\gamma_c(E_S) = N_S$. Given $n \in N_S$, choose a lift $w \in \gamma_c(H_S)$. Its base is 1, since $\gamma_c(B) = 1$. The same proper-support argument makes all coordinates at $T \not\subseteq S$ trivial. The remaining proper-support

coordinates lie in N_T; by induction their pure copies belong to $\gamma_c(H)$. Removing them leaves the pure n, still in $\gamma_c(H)$. All pure kernels are obtained, proving (2.6). □

Corollary U.SK.Separator (factorization across a separator). Under U.SK, let $I = L \cup M$. Suppose $B = B_L B_M$ as a product of sets, and every support in $\mathcal{F}$ is contained in L or in M. Then $H = H_L H_M$.

Proof. For $h \in H$, factor $\pi(h) = b_L b_M$ and choose lifts $u_L \in H_L$, $u_M \in H_M$. Then $h = n u_L u_M$ for some $n \in \ker \pi$. Assign every pure coordinate factor of n to one of the two sides containing its support. Formula (2.4) gives $n = n_L n_M$, with $n_L \in H_L$, $n_M \in H_M$. Each pure N_S is normal in H, because conjugation in the ambient product acts coordinatewise and $N_S \trianglelefteq E_S$. Hence the product of the assigned M-factors is normal in H. Moving n_M past u_L gives

$$h = (n_L u_L)(u_L^{-1} n_M u_L \, u_M) \in H_L H_M.$$

The opposite inclusion is automatic. □

The hypotheses of U.SK do not identify H with a universal amalgam of the E_S. Additional cross-relations and a presentation-recognition argument may be necessary. Nor can injectivity on every proper parabolic be replaced by injectivity on individual markers alone.

Proposition U.Fiber (two compatible split factors). Let A and B be marked groups on finite label sets L and M, and put $J = L \cap M$. Suppose their J-parabolics are identified with the same marked group C, and that $p_A : A \to C$ and $p_B : B \to C$ are deletions. Then

$$A \times_C B \cong (\ker p_A \times \ker p_B) \rtimes C. \tag{2.7}$$

This full fiber product is generated by the tuples $(a_i, 1)$ for $i \in L \setminus J$, $(1, b_i)$ for $i \in M \setminus J$, and (c_i, c_i) for $i \in J$. The two original groups embed through $a \mapsto (a, p_A(a))$ and $b \mapsto (p_B(b), b)$. If finite, its order is $|A||B|/|C|$.

Proof. Every compatible pair has a unique form $(n_A c, n_B c)$, giving the split decomposition and the order formula. The stated copies of A and B are generated by subsets of the stated tuple marking. They contain $(n_A, 1)$ and $(1, n_B)$, respectively, while either contains the diagonal copy of C. These subgroups generate the split product, proving the assertion. The embedding formulas are injective because their first and second projections, respectively, are identities. □

3. Finite two-group quotients with a fixed class-two shadow

The next theorem separates a universal finite search space from the task of proving that a particular family has the required class-two quotient. That identification is an application hypothesis, not a consequence of the search.

Theorem U.Nilpotent (complete marked kernel parameter space). Let $F = \langle x_i : i \in I \rangle$, where $|I| = d$ and $x_i^2 = 1$. Suppose F has all Boolean deletions. Put $Q = F/\gamma_3(F)$, with its image marking. For $k \geq 0$, set

$$U = F/\gamma_{k+3}(F), \qquad K = \gamma_3(U), \qquad K_S = K \cap U_S. \tag{3.1}$$

Then U is a finite two-group,

$$|U| \leq 2^{\sum_{j=1}^{k+2} d^j}, \tag{3.2}$$

the markings of U and Q have Boolean deletions and IP, and U/K is canonically the marked group Q. For every $S \subseteq I$,

$$K_S = \gamma_3(U_S). \tag{3.3}$$

There is a bijection between the following two sets:

1. Ordered marked isomorphism classes of finite two-group quotients $f : F \twoheadrightarrow G$ for which the induced marked map $Q \to G/\gamma_3(G)$ is an isomorphism, $|G| = |Q|2^k$, and G has IP.

2. Normal subgroups $N \trianglelefteq U$ satisfying $N \le K$, $[K : N] = 2^k$, and

$$\frac{NK_S}{N} \cap \frac{NK_T}{N} = \frac{NK_{S\cap T}}{N} \quad (S, T \subseteq I). \tag{3.4}$$

The correspondence sends N to U/N with the original image markers. If $D \le \mathrm{Sym}(I)$ is specified and each $\sigma \in D$ induces an automorphism $x_i \mapsto x_{\sigma(i)}$ of F, then equivalence allowing precisely these label permutations is the D-orbit relation on the kernels in item 2.

Proof. The degree-one part of the lower-central graded Lie ring of F is generated by elements of exponent dividing two. By multilinearity, every iterated commutator in those generators also has exponent dividing two in its homogeneous component. In degree j, at most d^j left-normed commutators suffice to generate that component. Consequently

$$|\gamma_j(F)/\gamma_{j+1}(F)| \le 2^{d^j}.$$

Multiplying these bounds for $1 \le j \le k+2$ proves finiteness, the two-power order and (3.2). Each deletion preserves every lower-central term, and therefore descends both to U and to Q. Proposition U.D gives IP for both image markings; no separate IP hypothesis on Q is needed. The epimorphic-image property of the lower central series gives $U/K \cong F/\gamma_3(F) = Q$.

For (3.3), one inclusion follows from $U_S \le U$. If $z \in \gamma_3(U) \cap U_S$, the deletion e_S fixes z and sends $\gamma_3(U)$ into $\gamma_3(U_S)$. This proves the reverse inclusion.

Let G satisfy item 1. Its subgroup $\gamma_3(G)$ has order 2^k. In a nilpotent group, equality between two successive nontrivial lower-central terms would persist at every later term, contradicting termination of that series. Thus each nontrivial step beginning at $\gamma_3(G)$ strictly decreases its order. There are at most k such steps, and $\gamma_{k+3}(G) = 1$. The map $F \to G$ therefore factors through U. Its kernel N lies in K, since the induced map $U/K \to G/\gamma_3(G)$ is an isomorphism, and the orders give $[K : N] = 2^k$.

Conversely, let $N \trianglelefteq U$ with $N \le K$ and $[K : N] = 2^k$, and write $G = U/N$. Its third lower-central subgroup is K/N, its canonical class-two quotient is Q, and its order is $|Q|2^k$. If $q : U \to G$, then

$$(K/N) \cap G_S = q(K \cap U_S) = NK_S/N. \tag{3.5}$$

Indeed, $q(u) \in K/N$ with $u \in U_S$ implies $u \in K$, because $q^{-1}(K/N) = K$. The reverse inclusion is immediate. Applying U.K to $G \to Q$ proves that IP in G is equivalent to precisely (3.4).

An isomorphism preserving each ordered marker commutes with the two maps from U, because it agrees with them on a generating family. Their kernels are consequently equal. Equal kernels plainly give the same marked quotient. Finally, every specified label permutation acts on F, preserves its characteristic lower-central terms, and hence acts on U and on the displayed kernels. A marker-permuting isomorphism induces the corresponding kernel equality after this automorphism, and the converse descends that automorphism to the quotients. This proves the orbit assertion at exactly the specified equivalence scope. □

Theorem U.Ladder (invariant index-two descent). Let U be a finite two-group and $M \trianglelefteq U$. The U-normal subgroups $L \le M$ of index two are exactly the inverse images of the hyperplanes of

$$M/(\Phi(M)[M, U]). \tag{3.6}$$

For $K \trianglelefteq U$, every U-normal subgroup $N \le K$ of index 2^k is obtained by k successive such steps.

Proof. An index-two subgroup L of M contains $\Phi(M)$ and has quotient C_2. If L is U-normal, conjugation by U on M/L is trivial, since $\mathrm{Aut}(C_2) = 1$. Thus $[M, U] \le L$. Conversely, a hyperplane inverse image has index two and contains $[M, U]$; for $l \in L$ and $u \in U$, the conjugate of l differs from l by an element of $[M, U]$, so lies in L. The quotient in (3.6) is elementary abelian by Burnside's basis theorem, proving the first assertion.

Suppose $N \trianglelefteq U$ and $N < M$. The nontrivial finite two-group M/N has a nonzero Frattini quotient V. Conjugation by U/N acts on the nonzero linear maps $V \to \mathbb{F}_2$. If $\dim V = a$, there are $2^a - 1$ such maps, an odd number. Every orbit has two-power size, so at least one orbit has size one. The kernel of a fixed nonzero map lifts to a U-normal subgroup between N and M, of index two in M. Repeating with the new subgroup gives a chain terminating at N; its length is k, since the total index is 2^k. □

Lemma U.Order (presentation recognition by a sharp bound). Let $A_1, \dots, A_t$ be finite normal subgroups of G, with $|A_j| \le a_j$. Suppose $[G : A_1 \cdots A_t] \le b$. Then $|G| \le b \prod_j a_j$. If a finite quotient of G has this latter order, its quotient map is an isomorphism.

Proof. Normality makes each partial product a subgroup. The multiplication map from $A_1 \times \cdots \times A_t$ onto the product is surjective as a map of sets, so that product has order at most $\prod_j a_j$. Multiplication by the quotient index proves the bound and finiteness of G. A quotient attaining the bound forces equality of the finite orders and hence a trivial kernel. □

4. Coset comparison and coordinate-kernel geometry

For subgroups $P_i \le G$, typed coset Helly means that every pairwise intersecting family of right cosets $P_i g_i$, with distinct indices, has a common element. This property is separate from IP. It supplies a common representative for an incident flag, whereas IP determines the stabilizer of a flag already represented in this way.

Lemma U.Helly (anchored criterion). A finite indexed subgroup family $(P_i)_{i \in I}$ has typed coset Helly if and only if the following condition holds. For every $i \in I$, every $J \subseteq I \setminus \{i\}$ with $|J| \ge 2$, and any selected $j_0 \in J$, each $g \in P_{j_0}$ satisfying $g \in P_i P_j$ for all $j \in J$ admits $k \in \bigcap_{j \in J} P_j$ with $P_i g = P_i k$.

Proof. The cosets $P_i g$ and P_j intersect precisely when $g \in P_i P_j$. Thus Helly gives a common element k, proving necessity. Conversely, induct on the number of cosets in a pairwise intersecting family. A family of at most two already has a common element. For a larger family, the induction hypothesis gives a common element of all but one coset. Right multiplication by its inverse turns those cosets into the base cosets P_j. The remaining coset meets P_{j_0}, so it has a representative $g \in P_{j_0}$. Its intersections with the other base cosets are precisely the product conditions in the statement. The criterion supplies k, and undoing the right translation gives a common element of the original family. □

Theorem U.Compare (all-parabolic comparison). Let $G = \langle r_i : i \in I \rangle$, where I is finite and $|I| \ge 3$. Put $P_i = G_{I \setminus \{i\}}$, fix $e \in I$, and set

$$\mathcal{B} = \bigcup_{i \ne e} P_i, \qquad \mathcal{W} = \mathcal{B}\mathcal{B}. \tag{4.1}$$

Let $q : G \twoheadrightarrow H$ be a marked epimorphism. Suppose q is injective on the set $\mathcal{W}$, and, for every $w \in \mathcal{W}$,

$$q(w) \in q(P_e) \quad \Longleftrightarrow \quad w \in P_e. \tag{4.2}$$

Then the marking of G has IP if and only if that of H has IP. Independently, the maximal-parabolic cosets of G have typed coset Helly if and only if those of H do.

Proof. The set $\mathcal{B}$ contains 1, is closed under inverses, and is contained in $\mathcal{W}$. If $S \subsetneq I$ and $S \ne I \setminus \{e\}$, then S omits a nonexceptional index, so $G_S \subseteq \mathcal{B}$. It follows that, for $b \in \mathcal{B}$ and every proper S,

$$q(b) \in q(G_S) \quad \Longleftrightarrow \quad b \in G_S. \tag{4.3}$$

For the exceptional index set this is (4.2). Otherwise, lift the image membership to $a \in G_S \subseteq \mathcal{B}$; injectivity on $\mathcal{W}$ and $q(a) = q(b)$ imply $a = b$.

For two proper index sets S, T, at least one corresponding subgroup is contained in $\mathcal{B}$, unless both index sets equal $I \setminus \{e\}$. In the former case, lift an element of $q(G_S) \cap q(G_T)$ through that protected

subgroup, and use (4.3) to put the lift in the other subgroup. Thus q identifies the actual intersection bijectively with the intersection of the images, and also identifies $G_{S\cap T}$ with its image. Equality of these two subgroups is equivalent upstairs and downstairs. The remaining self-intersection is tautological, as is any pair involving $G_I = G$. This proves the IP equivalence without making any assumption about distinct index sets generating distinct subgroups.

The same argument gives two facts for Helly. First, whenever J contains a protected index $j_0 \neq e$, restriction of q is a bijection

$$\bigcap_{j\in J} P_j \longrightarrow \bigcap_{j\in J} q(P_j). \tag{4.4}$$

Lift through P_{j_0} and apply (4.3) for each remaining membership. Second, if $g \in \mathcal{B}$ and $i \neq j$, then

$$q(g) \in q(P_i)q(P_j) \quad \Longleftrightarrow \quad g \in P_iP_j. \tag{4.5}$$

When both indices are protected, a factorization downstairs lifts to $ab \in \mathcal{W}$; comparison with $g \in \mathcal{W}$ gives equality. When $i = e$, choose the protected second factor $b \in P_j$. The remaining membership test is $q(gb^{-1}) \in q(P_e)$, and $gb^{-1} \in \mathcal{W}$, so (4.2) applies. When $j = e$, multiply by the inverse of a protected first factor on the left. The reverse implication in all cases follows simply by applying q.

Apply U.Helly in the two groups. Every set J appearing there has at least two indices, hence contains a protected j_0. Every candidate representative in $q(P_{j_0})$ lifts uniquely to $g \in P_{j_0}$. Equation (4.5) preserves all its eligibility conditions, while (4.4) identifies the possible common representatives k. Finally,

$$P_ig = P_ik \quad \Longleftrightarrow \quad gk^{-1} \in P_i.$$

Here $g, k \in P_{j_0}$, so $gk^{-1} \in \mathcal{B}$, and (4.3) preserves this final test. Thus the anchored criterion has the same truth value upstairs and downstairs, proving the Helly equivalence. □

Corollary U.Compare.Matrix (bounded integral certificate). Suppose $G \leq \mathrm{GL}_m(\mathbb{Z})$, and let q_n be reduction modulo n. If all entries of all matrices in $\mathcal{W}$ have absolute value at most C, then q_n is injective on $\mathcal{W}$ for $n > 2C$. Suppose also that, for every $w \in \mathcal{W}$, a finite specified collection of integer tests has entries bounded by D, membership in $q_n(P_e)$ forces at least one complete test to vanish modulo n, and every integral passing case has a proved witness placing w in P_e. Then (4.2) holds for $n > D$.

Proof. Equal reductions of two matrices in $\mathcal{W}$ make every entry of their difference divisible by n and bounded by $2C$, hence zero. Similarly, a test vanishing modulo $n > D$ vanishes integrally. Its witness gives the forward implication in (4.2); the reverse implication is automatic. □

If several quotient maps satisfy U.Compare against the same source, no maps between the quotient groups are needed. One checked quotient determines IP and Helly in the source and in every such quotient. A finite protected product set still requires an exact membership test against the whole exceptional image, not a truncated search.

Proposition U.Geometry (coset recognition and faithful chamber action). Suppose the marking of G consists of nonidentity involutions, has IP, and has typed coset Helly for its maximal parabolics. The coset incidence system is then a thin, residually connected geometry, and G is its full type-preserving automorphism group, acting regularly on chambers. More generally, the full type-preserving automorphism group of any thin, residually connected geometry of finite rank acts freely on chambers. A chamber-transitive subgroup is consequently the full group.

Proof. In the coset system, a common coset point extends every flag to a chamber by choosing the coset of each missing type through that point. Right translation sends every such flag to its base flag. For a base flag of type set S, IP gives stabilizer $G_{I\setminus S}$. In particular the full chamber stabilizer is 1, while a panel stabilizer is a group generated by one nonidentity involution and has order two. There are therefore exactly two chambers through each panel.

Consider a residue with remaining type set T, where $|T| \geq 2$. Its stabilizer is G_T, and its maximal object stabilizers are $G_{T\setminus\{i\}}$, $i \in T$. Every marker indexed by T lies in at least one of these proper parabolics. A word in those subgroups supplies a walk between their translated base objects in the residue incidence graph; all objects are translates of base objects and all base objects lie in a common flag. The residue is therefore connected.

For the general assertion, first observe that every residually connected finite-rank geometry has a connected chamber graph. This follows by induction on rank: chambers through a fixed object are connected through galleries in its residue, and a path in the incidence graph, extending each consecutive incident pair to a chamber, connects arbitrary chambers through these subgraphs. The rank-one case holds because all chambers share the empty panel. In a thin geometry an automorphism fixing a chamber fixes its unique neighbor across each panel. It therefore fixes every chamber along every gallery, and hence every object. The chamber action is free. If a subgroup is chamber-transitive, any full automorphism agrees on a base chamber with one subgroup element; their quotient fixes that chamber and is the identity. Apply this to the coset geometry to obtain the stated full automorphism group. □

4.1. Helly transfer through coordinate kernels

Write right cosets as Pg, with the subgroup on the left. Let $q : H \to B$ be an epimorphism of groups generated by corresponding marked families $(h_i)_{i\in I}$ and $(b_i)_{i\in I}$, where I is finite. For $J \subseteq I$, write $H_J = \langle h_j : j \in J\rangle$ and $B_J = \langle b_j : j \in J\rangle$. In particular, $q(H_J) = B_J$ for every J.

Fix any set $\mathcal{J}$ of subsets of I as the permitted types. The maximal-parabolic case is $\mathcal{J} = \{I\setminus\{i\} : i \in I\}$.

Theorem U.SK.Helly (coordinate-kernel transfer). Suppose B has IP. Suppose $N = \ker q$ is an internal direct product $\prod_{\alpha\in A} N_\alpha$, with a finite factor set A, and for every $J \in \mathcal{J}$,

$$N \cap H_J = \prod_{\alpha\in A_J} N_\alpha \qquad \text{for some } A_J \subseteq A. \tag{4.6}$$

If the family $(B_J)_{J\in\mathcal{J}}$ has typed coset Helly, then so does $(H_J)_{J\in\mathcal{J}}$.

No commutativity within the factors, finiteness of H or B, or faithfulness of an action of B on N is assumed. The required intersection-image fact is $q(H_{J\cap K}) = B_J \cap B_K$; it follows here from the marked map and IP of B. IP of H is not an additional assumption for this transfer theorem.

Proof. Consider pairwise intersecting cosets $H_{J_i}g_i$. Their images $B_{J_i}q(g_i)$ intersect pairwise. Choose a common base element b by the assumed Helly property, and choose u in H with $q(u) = b$. Each set

$$F_i = H_{J_i}g_i \cap q^{-1}(b)$$

is nonempty: because $bq(g_i)^{-1} \in B_{J_i}$, a preimage of that element in H_{J_i}, multiplied on the right by g_i, belongs to F_i.

We first show that $F_i \cap F_j \neq \emptyset$. Choose $x \in H_{J_i}g_i \cap H_{J_j}g_j$ and put z=q(x). Both b and z belong to each of the two base cosets, so

$$bz^{-1} \in B_{J_i} \cap B_{J_j} = B_{J_i\cap J_j}.$$

The marked map is surjective on this intersection parabolic. Choose $h \in H_{J_i\cap J_j}$ with $q(h) = bz^{-1}$. Since h belongs to both H_{J_i} and H_{J_j}, left multiplication by h preserves both right cosets. Thus hx belongs to their intersection and has image b. This proves the assertion without assuming that q carries an arbitrary intersection onto the intersection of its images.

Choose $x_i \in F_i$ and put $k_i = x_iu^{-1} \in N$. If $y \in F_i$, then $yx_i^{-1} \in N \cap H_{J_i}$; the converse also holds. Hence

$$F_iu^{-1} = (N \cap H_{J_i})k_i. \tag{4.7}$$

Under the direct product coordinates of N, (4.6) makes (4.7) unrestricted in coordinates $\alpha \in A_{J_i}$, and fixes coordinate $\alpha \notin A_{J_i}$ to $(k_i)_\alpha$. Pairwise intersections imply that any two constraints fixing the same coordinate fix it to the same value. Choose that value in every constrained coordinate and the identity in every unconstrained coordinate. The resulting element belongs to all sets in (4.7). Multiplication on the right by u supplies a common element of the original cosets. This proves the theorem. □

Corollary U.SK.Geometry. In the supported-kernel realization U.SK, assume the base marking gives a regular hypertope and every marked generator in every attached group is an involution. Then the resulting marking also gives a regular hypertope, and its full type-preserving automorphism group is the constructed group.

Proof. U.SK gives IP and the exact profile

$$N \cap H_J = \prod_{S \subseteq J} N_S.$$

The base maximal-parabolic cosets have typed Helly. Theorem U.SK.Helly applies with factors indexed by the supports S. Each constructed generator is a tuple whose base coordinate and every nontrivial attached coordinate are involutions; hence its square is the identity. Its base image is nonidentity, so its order is exactly two. U.Geometry now gives thinness, residual connectedness, chamber regularity and equality with the full type-preserving automorphism group. In particular, this conclusion uses typed Helly in addition to IP. □

A sufficient alternative to local involutivity is that every support has size at least two: each singleton local parabolic is then proper, and injectivity into its nonidentity involutory base image forces the local marker to have order two. This is a sufficient condition, not an equivalent reformulation. Without either condition the geometric corollary fails. With a singleton type set and support, the quotient $C_4 \to C_2$ is injective on its only proper parabolic, the trivial subgroup. U.SK then constructs C_4, whose distinguished generator has order four. In rank three, adjoining two independent C_2 markers gives $C_4 \times C_2^2$ with the same defective singleton attachment and a four-element panel.

5. Retained objects and orbit multiplicities

The representation of a group by permutation matrices is exact: its regular action is faithful, and replacing permutations by their matrices preserves multiplication. For geometric applications one must also retain the distinguished parabolic partitions. A matrix realization alone does not establish the intersection or common-point properties of those partitions.

Theorem U.Action (faithful action after retaining objects). Let G act on sets X and Y. Suppose a G-invariant subset $Y_0 \subseteq Y$ has an equivariant surjection onto X. Suppose also that Y has an injective equivariant encoding in a tagged disjoint union of finite products, each factor of which is X or its power set, and that the tags are fixed by G. Then

$$\ker(G \curvearrowright Y) = \ker(G \curvearrowright X) = \bigcap_{x \in X} G_x. \tag{5.1}$$

If X is a disjoint union of transitive orbits with selected stabilizers H_j, the common kernel is

$$N = \bigcap_j \operatorname{core}_G(H_j) = \operatorname{core}_G\left(\bigcap_j H_j\right). \tag{5.2}$$

The faithful permutation image is G/N. If its action on Y is chamber-transitive on a thin, residually connected finite-rank geometry, then G/N is the full type-preserving automorphism group. The preimages in G of a chamber stabilizer and of a panel stabilizer are respectively N and a subgroup of index two over N.

Proof. An element fixing Y pointwise fixes X through the equivariant surjection. Conversely, an element fixing X fixes all subsets of X, every finite tuple formed from these objects, and all fixed tags. Injectivity of the encoding then forces it to fix Y pointwise. This proves (5.1). In a transitive orbit all point stabilizers are conjugates of the selected one, so intersection over all points gives the first equality in (5.2). The second is the intersection of the same conjugate subgroups in the opposite order. The first isomorphism theorem identifies the faithful image.

By U.Geometry that image has trivial chamber stabilizers and is the full type-preserving group. Its panel stabilizer acts transitively on the two chambers through the panel, with trivial stabilizer of either chamber, so has order two. Taking preimages proves the final assertion. □

Independently labeled copies of the same subset need extra data in their encoding; they cannot be silently treated as the subset itself.

Theorem U.Orbits (orbit data and normalizers). Let a finite group A act on a finite set Ω, and choose one representative H of each conjugacy class of subgroups that occurs as a point stabilizer. Let m_H be the number of orbits isomorphic to A/H, with $m_H = 0$ for all omitted classes. Then the numbers m_H determine the A-set up to equivariant isomorphism, and

$$\ker(A \curvearrowright \Omega) = \bigcap_{m_H>0} \operatorname{core}_A(H). \tag{5.3}$$

Its equivariant permutation group is abstractly

$$C_{\mathrm{Sym}(\Omega)}(A) \cong \prod_H ((N_A(H)/H) \wr \mathrm{Sym}(m_H)). \tag{5.4}$$

If the action $\rho : A \hookrightarrow \mathrm{Sym}(\Omega)$ is faithful, conjugation gives an exact sequence

$$1 \longrightarrow C_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow N_{\mathrm{Sym}(\Omega)}(\rho(A)) \longrightarrow \{\alpha \in \mathrm{Aut}(A) : m_{\alpha(H)} = m_H \text{ for all } H\} \longrightarrow 1. \tag{5.5}$$

No splitting of this sequence is asserted.

Proof. Every orbit is equivariantly A/H, where H is the stabilizer of a selected point, and two such transitive sets are isomorphic exactly when the stabilizers are conjugate. This proves the classification by multiplicities and the kernel formula.

An equivariant permutation preserves the isomorphism type of each orbit. For one orbit A/H, an equivariant map is determined by the image of H. That image nH must be fixed by H, and bijectivity makes its stabilizer equal to H; hence $n \in N_A(H)$. Conversely each such n gives a permutation $aH \mapsto anH$. Composition corresponds to multiplication in the opposite order, so inversion identifies this permutation group with $N_A(H)/H$. For m_H copies, one may independently choose such a map in every copy and permute the copies, yielding the wreath product. Different isomorphism types are independent, proving (5.4).

Faithfulness makes conjugation by a normalizing permutation τ induce a unique automorphism α of A, with

$$\tau\rho(a)\tau^{-1} = \rho(\alpha(a)).$$

If a point has stabilizer H, its image under τ has stabilizer $\alpha(H)$; therefore the multiplicities must be invariant. The kernel of this conjugation map is the centralizer. Conversely, suppose α preserves the multiplicities. Pair each orbit with a target orbit whose stabilizer class is $\alpha(H)$, and choose base points x, y with actual stabilizers $H, \alpha(H)$, adjusting within the target orbit if needed. Define $\tau(ax) = \alpha(a)y$. The stabilizer equality makes this well-defined and bijective on that pair of orbits, and the paired definitions combine to a permutation of Ω. It satisfies the displayed intertwining equation, proving surjectivity and exactness. □

Proposition U.ExtensionOrbits (classification above a retained base). Let a group B act on sets Z, E, and let $p : Z \to E$ be equivariant. For one representative e of each orbit in $B\backslash E$, there is a natural bijection

$$B\backslash Z \;\cong\; \coprod_{e\in B\backslash E} B_e\backslash p^{-1}(e). \tag{5.6}$$

Proof. Translate any $z \in Z$ so that $p(z)$ is the selected representative of its orbit. If two translated points over that representative are B-equivalent, an element taking one to the other fixes their image e, so belongs to B_e. Conversely a B_e-equivalence is already a B-equivalence. These two observations prove surjectivity and injectivity of (5.6). □

6. Relative generation and Frattini coordinates

A finite indexed family $(s_i)_{i\in I}$ is independent relative to $B \le G$ if

$$s_i \notin \langle B, s_j : j \neq i\rangle \quad (i \in I).$$

It is a relative generating family if it also generates G together with B. For a finite group G, let $m(G)$ be the largest size of an irredundant generating family and let $i(G)$ be the largest size of an independent family, without the generation requirement. If G acts on a subgroup N, independence under this action means that no member is generated by all G-conjugates of the other members; write $i_G(N)$ for the corresponding maximum.

Proposition U.RelativeBound (normal quotient bound). If G is finite and $N \trianglelefteq G$, then

$$m(G) \le m(G/N) + i(N). \tag{6.1}$$

If N is abelian, then $i(N)$ can be replaced by $i_G(N)$.

Proof. Take an irredundant generating family S of G, and a subfamily $T \subseteq S$ whose images minimally generate G/N. For every $s \in S\backslash T$, choose a word w_s in T with $sN = w_sN$, and put $v_s = sw_s^{-1} \in N$. If one v_s belonged to the subgroup generated by the other v_t, substituting $v_t = tw_t^{-1}$ would express s using only $S \setminus \{s\}$. Thus the v_s are independent. Since $|T| \le m(G/N)$, (6.1) follows.

If N is abelian, the equality $G = N\langle T\rangle$ says that conjugation by G on N is already conjugation by $\langle T\rangle$. An expression for v_s using conjugates of the other v_t would again substitute to an expression for s without s. The family is therefore G-independent, proving the sharper bound. □

For a finite indexed subgroup family $H_i \le G$, write $H_J = \bigcap_{j\in J} H_j$, with $H_\emptyset = G$ and $B = H_I$.

Theorem U.Relative (generation from residual maximality). Suppose all H_J, $J \subseteq I$, are distinct, and for every $J \subsetneq I$ there is $k \notin J$ such that $H_{J\cup\{k\}}$ is maximal in H_J. Choose

$$s_i \in H_{I\setminus\{i\}} \setminus H_i. \tag{6.2}$$

Then

$$H_J = \langle B, s_i : i \notin J\rangle \quad (J \subseteq I). \tag{6.3}$$

In particular these elements generate G irredundantly relative to B, and their relative parabolics satisfy

$$\langle B, s_i : i \in S\rangle \cap \langle B, s_i : i \in T\rangle = \langle B, s_i : i \in S \cap T\rangle. \tag{6.4}$$

The group need not be finite.

Proof. Distinctness of $H_{I\setminus\{i\}}$ and B makes (6.2) possible, since its intersection with H_i is B. Every s_j with $j \neq i$, and every element of B, belongs to H_i, while s_i does not. This proves relative independence.

Prove (6.3) by downward induction from $J = I$. Choose k as in the hypothesis. By induction, $H_{J\cup\{k\}}$ is generated by B and the s_i with $i \notin J \cup \{k\}$. The element s_k belongs to H_J, because $J \subseteq I \setminus \{k\}$, and lies outside $H_{J\cup\{k\}}$, because it lies outside H_k. Maximality therefore says that adjoining s_k generates H_J. This is (6.3). Finally substitute $J = I\setminus S$ and $I\setminus T$ into the elementary equality $H_J \cap H_{J'} = H_{J\cup J'}$ to obtain (6.4). □

Lemma U.Maximal (relative finite generation). If $G = \langle B, t_1, \dots, t_r\rangle$ with r finite, then every proper subgroup containing B is contained in a maximal subgroup of G.

Proof. Order the proper overgroups of the given subgroup by inclusion. The union of a chain is a subgroup. If that union were G, a single member of the chain would contain all the finitely many t_j, and hence all of G, a contradiction. Thus every chain has a proper upper bound. Zorn's lemma gives a maximal proper overgroup. □

Define $\Phi(G)$ to be the intersection of all maximal subgroups, with intersection G if there are none. This subgroup is characteristic. Whenever every proper overgroup of B is contained in a maximal subgroup, a subgroup $H \geq B$ satisfies

$$\Phi(G)H = G \quad \Longleftrightarrow \quad H = G : \tag{6.5}$$

a maximal overgroup of a proper H would contain both factors on the left.

Theorem U.FrattiniTransport (relative generating families). Under the maximal-overgroup hypothesis of (6.5), a finite indexed family is an irredundant relative generating family of G over B if and only if its image is one of $G/\Phi(G)$ over $B\Phi(G)/\Phi(G)$.

Proof. Generation in the quotient says $\Phi(G)\langle B, s_i : i \in I\rangle = G$, which is equivalent to generation in G by (6.5). Once generation holds, irredundancy says that each deletion subgroup $\langle B, s_j : j \neq i\rangle$ is proper. Equation (6.5), applied to each of those subgroups, says exactly that its image is proper in the quotient. These are precisely the quotient irredundancy conditions. □

The maximal-overgroup hypothesis holds for finite G and, by U.Maximal, for relative finite generation. It cannot simply be omitted for arbitrary infinite groups: in $C_{p^\infty} \times C_2$, the divisible factor is the Frattini subgroup, and a generator of the quotient does not generate the whole group.

Theorem U.FrattiniLadder (containment in the common stabilizer). Suppose the H_J are all distinct and some ordering of $I = \{1, \dots, r\}$ gives a chain

$$G = K_0 > K_1 > \cdots > K_r = B, \qquad K_j = H_{\{1,\dots,j\}}, \tag{6.6}$$

in which every inclusion is maximal. If $N \trianglelefteq G$ lies in every maximal subgroup of G, then $N \leq B$.

Proof. Choose $t_j \in K_{j-1} \setminus K_j$. Maximality gives $K_{j-1} = \langle K_j, t_j\rangle$, and iteration gives $G = \langle B, t_1, \dots, t_r\rangle$. Every proper H_i therefore lies in a maximal subgroup by U.Maximal. Such a maximal subgroup contains N, so $NH_i \neq G$.

Suppose $N \not\leq B$, and choose the first k with $N \not\leq K_k$. Then $N \leq K_{k-1}$, and normality and maximality give

$$K_{k-1} = N(K_{k-1} \cap H_k).$$

Suppose at an upward step that $K_j = N(K_j \cap H_k)$, where $1 \leq j < k$. Since $N \leq K_{j-1}$,

$$K_j \leq N(K_{j-1} \cap H_k) \leq K_{j-1}.$$

If this middle group were K_j, then $K_{j-1} \cap H_k \leq K_j$, forcing

$$K_{j-1} \cap H_k = K_j \cap H_k.$$

The corresponding index sets differ because $j < k$, contradicting the assumed distinctness. Maximality therefore makes the middle group K_{j-1}. Continuing upward reaches $G = NH_k$, contradicting the first paragraph. Hence $N \leq B$. □

Theorem U.Phi (primitive saturation and coordinates). Let P be a finite p-group with an irredundant generating marking $(r_i)_{i\in I}$, $|I| = d$, and let $F = \Phi(P)$. Then, for every $S \subseteq I$,

$$\Phi(P_S) = P_S \cap F, \qquad P_S F = \pi^{-1}\,\mathrm{span}(\bar{r}_i : i \in S), \qquad [P : P_S F] = p^{d-|S|}. \tag{6.7}$$

The saturated parabolics $P_S F$ intersect according to intersection of index sets. Each omission subgroup $P_{I\setminus\{i\}}$ has exactly one maximal overgroup, namely $P_{I\setminus\{i\}}F$.

If $B \geq F$ and $(s_i)_{i\in J}$ is an irredundant relative generating family over B, then $B \trianglelefteq P$, P/B has basis $(s_i B)_{i\in J}$ over $\mathbb{F}_p$, and

$$\langle B, s_i : i \in T\rangle/B = \mathrm{span}(s_i B : i \in T). \tag{6.8}$$

If $p = 2$ and all original r_i are involutions, then $F = P'$ and $\Phi(P_S) = P'_S$ for every S.

Proof. Burnside's basis theorem makes the $\bar{r}_i$ a basis of P/F. The image of P_S therefore has order $p^{|S|}$, with kernel $P_S \cap F$. Since this image is elementary abelian, $\Phi(P_S) \leq P_S \cap F$. On the other hand P_S is generated by $|S|$ elements, so Burnside gives $[P_S : \Phi(P_S)] \leq p^{|S|}$. The quotient by $P_S \cap F$ already has that order. Equality follows, proving the first formula. The remaining formulas and the intersection assertion are inverse-image statements for coordinate subspaces.

Every maximal subgroup of a finite p-group contains F and corresponds to a hyperplane of P/F. A hyperplane containing a specified coordinate hyperplane equals it, proving uniqueness of the maximal overgroup of an omission subgroup.

For the relative assertion, every subgroup containing F is normal, because P/F is abelian. The quotient P/B is elementary abelian. The indicated images span it, and any nontrivial linear relation would express one s_i using B and the other markers, contradicting irredundancy. They form a basis, proving (6.8). Finally, an abelian group generated by involutions is elementary abelian of exponent two. Thus $P' \leq \Phi(P)$, while the elementary abelian quotient P/P' gives the reverse inclusion. The same argument applies to each P_S. □

These coordinates also describe the corresponding omission cosets: a coset prescribes one coordinate value, and every simultaneous assignment gives a chamber. The geometry is complete multipartite, with p objects in each type, and the quotient translation group acts regularly on chambers. For $p > 2$ it need not be the full type-preserving automorphism group; the thin case $p = 2$ is the one covered by U.Geometry.

7. How the statements are used

The coordinate-kernel theorem U.SK.Helly additionally requires exact kernel profiles and base IP to move pairwise intersections to a common fiber. Its geometric corollary requires actual involutory local markers, as well as a regular base geometry. These obligations are checked for the three dihedral pair attachments in Module A.

The application interface is mathematical rather than merely notational. An application of U.SK supplies the marked base, every actual deletion, each local marked epimorphism, and injectivity on every proper local parabolic. Its order formula then needs only the orders of the base and local kernels. Its universal-presentation claim, if any, still requires a presentation map and an independent sharp bound such as U.Order. An application of U.Nilpotent must first identify the canonical class-two quotient of every target group. An application of U.Compare must prove injectivity on the whole protected product set and exact membership in the exceptional image. An application of U.Action supplies the actual encoding and retained surjection, not an assumed equality of kernels.

These requirements permit the same complete proof to be cited several times without repeating its induction or quotient argument. They also identify the portions that cannot be omitted from an application: local models, presentation recognition, geometric realization, source-specific parameter ranges, and any

exhaustive finite computation on which its result depends. The companion contract ledger records these bindings separately from the historical attribution of each original theorem.

8. Genuine group pushouts and primary torsion

Fix injective homomorphisms $i_A : C \to A$ and $i_B : C \to B$. Let $A * B$ be the free product and let

$$P = (A * B)/\langle\langle i_A(c) i_B(c)^{-1} : c \in C \rangle\rangle. \tag{8.1}$$

We use the amalgam normal-form theorem in the following precise form: the canonical maps of A and B into P are injective; an alternating reduced word with factors outside the common copy of C has nonzero reduced length and is nonidentity; every element is conjugate either into a factor or to a cyclically reduced alternating word of length at least two, whose positive powers remain reduced with proportionally increasing length. This is a retained classical theorem.

Theorem U.Amalgam (universal property and torsion). The group P in (8.1), with its canonical maps j_A, j_B, is a pushout of $A \leftarrow C \to B$ in the category of groups: for every group D and homomorphisms $f_A : A \to D$, $f_B : B \to D$ satisfying $f_A i_A = f_B i_B$, there is a unique homomorphism $f : P \to D$ with $f j_A = f_A$ and $f j_B = f_B$.

Let p be a prime. Say that a group has p-primary finite torsion when every finite-order element has p-power order. This property passes to subgroups, to extensions with finite p-group quotient, and to amalgamated free products of groups having the property. If $i_A(C) \neq A$ and $i_B(C) \neq B$, then P is infinite. If A and B are finitely generated, then so is P.

Proof. The free-product universal property gives a unique homomorphism $A * B \to D$ extending f_A, f_B. Agreement on C puts every defining relator in its kernel, so it factors through P. The images of A and B generate P; hence this factorization is unique. The equality $j_A i_A = j_B i_B$ follows from the defining relations, completing the pushout property.

The torsion assertion for subgroups is immediate. For an extension $1 \to K \to E \to R \to 1$, with R a finite p-group and K having p-primary finite torsion, take any finite cyclic subgroup $\langle x \rangle$ of E. Its intersection with K is a cyclic p-group, and its image in R is a p-group. The order formula for this restricted exact sequence makes $|x|$ a power of p.

For an amalgam, the normal-form theorem places a torsion element, up to conjugacy, in a factor: the alternative cyclically reduced word has powers of unbounded reduced length and cannot have finite order. The factor hypothesis therefore proves the torsion assertion. If the common subgroup is proper in both factors, choose $a \in A \setminus i_A(C)$ and $b \in B \setminus i_B(C)$. The words $(ab)^n$ are reduced of length $2n$, so are all distinct; this proves infinitude. Finally, the images of finite generating sets of A and B generate their free product and its quotient P, proving finite generation. □

The simultaneous fiber product of U.SK has its displayed compatibility equations and its proved generation theorem. It is not the pushout in U.Amalgam. A graph of proof dependencies is neither of these group constructions.

Attribution and references

The elementary image, kernel, orbit and amalgam arguments above use established group theory. The relative-generation and Frattini applications have their source context in Cameron–Cara [CC02] and Archer–Cara–Krempa [ACK05]; their original results retain that attribution. The constructions motivating the supported-kernel and finite-quotient interfaces are those of [HFLa], [HFLb], [HS26] and [CFHS25], together with the order-1024 study [GLP18] and the three-generator hypertope construction [HFLh]. Their source-specific results are stated and credited in Application A, where the canonical forest quotient, square model, classification branches and computations are proved or expressly retained at their actual scope.

Within this project, U.K, U.Q, U.D, U.Compare, U.Action and the relative and Frattini arguments reuse earlier written derivations rather than starting their proofs again. U.SK abstracts the common supported-extension construction. Its inclusion-induction proof removes the earlier separate normal-generator-word hypothesis, and its deletion corollary derives a hypothesis that had previously been assumed. U.Nilpotent separates the previously forest-specific search theorem into a general exact-quotient statement and an application-specific proof identifying that quotient. These descriptions concern the organization and development of the present proof library; they are not claims of priority over the literature. The separate provenance ledger binds the exact predecessor proofs, the new interfaces and the limited existing Lean certificates. The coordinate-kernel Helly proof was developed as the common geometric step for the three-generator hypertope construction and is integrated here with its necessary local-involutivity hypothesis. The affine and source-presentation arguments remain in Module A. These additions make no historical novelty or open-conjecture claim.

[CC02] P. J. Cameron and P. Cara, *Independent generating sets and geometries for symmetric groups*, Journal of Algebra **258** (2002), 641–650. DOI: 10.1016/S0021-8693(02)00550-1. The retained target edition is the twelve-page author manuscript.

[ACK05] C. Archer, P. Cara and J. Krempa, *Using the Frattini Subgroup and Independent Generating Sets to Study RWPRI Geometries*, Beiträge zur Algebra und Geometrie **46** (2005), 169–177. The retained target edition is the published nine-page text.

[HFLa] D.-D. Hou, Y.-Q. Feng and D. Leemans, *On regular polytopes of 2-power order*, revised author manuscript dated 8 November 2019; journal publication 2020. DOI: 10.1007/s00454-019-00119-5.

[HFLb] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-polytopes of order* 2^n, arXiv:1802.08947v2, 22 January 2019; journal title *2-groups behaving as automorphism groups of regular 3-polytopes*.

[HS26] D.-D. Hou and E. Schulte, *String C-groups of 2-power order project onto a common string C-group*, arXiv:2605.20691v1, 20 May 2026.

[CFHS25] G. Cunningham, Y.-Q. Feng, D.-D. Hou and E. Schulte, *Symmetric polytopes whose automorphism groups are 2-groups*, arXiv:2512.15511v1, 17 December 2025.

[GLP18] Y. Gomi, M. L. Loyola and M. L. A. N. De Las Peñas, *String C-groups of order 1024*, Contributions to Discrete Mathematics **13**(1) (2018), 1–22. DOI: 10.55016/ojs/cdm.v13i1.62475. The exact published edition and presentation corrections are specified in Application A's source contract; this umbrella does not claim to reproduce that census on its own.

[HFLh] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-hypertopes with* 2^n *chambers*, arXiv:1901.09034v1, 23 January 2019. The exact presentation and corrected structural displays are treated in Module A.